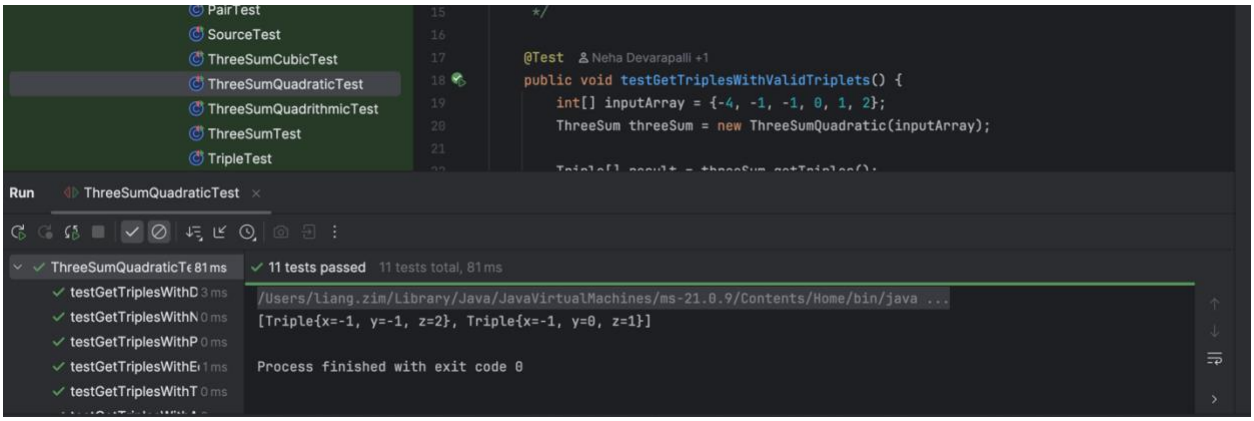
N	Cubic (ms)	Quadrithmic (ms)	Quadratic (ms)
250	9	4	3
500	23	6	3
1000	148	12	7
2000	1077	54	37
4000	8483	209	201



The Quadratic method reduces the time complexity from $O(n^3)$ to $O(n^2)$. a[j] is already fixed, and i and k move from both sides toward the middle. If sum < 0, the sum is too small, move the left pointer right; if sum > 0, the sum is too large, move the right pointer left; if sum == 0, the target is found. This way, the third loop is not needed. The Quadrithmic method has a time complexity of $O(n^2 \log n)$, requires log n comparison operations, and is also slower than Quadratic.